Basic HTML and CSS Website Demo
KHE Hackathon 2025

Setup Instructions
1. Create a new folder on your computer called "my-website"
2. Inside this folder, create two files:
   - index.html
   - styles.css
3. Open both files in your code editor (like VS Code, Sublime Text, or Notepad++)

Part 1: Understanding the HTML Structure
The HTML file (index.html) is divided into several key sections:

1. Document Setup
- The DOCTYPE declaration tells browsers this is an HTML5 document
- The `<html>` tag contains language information
- The `<head>` section contains metadata and links to CSS
- The `<body>` contains all visible content

2. Header Section
- Contains the main title of the website
- Styled with a dark background and white text

3. Navigation Menu
- Created using an unordered list (`<ul>`)
- Contains links to different sections
- Styled to appear as a horizontal menu

4. Main Content
- Wrapped in a container for centered layout
- Contains articles with headings, paragraphs, and buttons
- Includes a placeholder image

5. Footer
- Contains copyright information
- Styled similarly to the header

Part 2: Understanding the CSS Structure
The CSS file (styles.css) is organized by sections:

1. CSS Reset
- Removes default browser styling
- Ensures consistent appearance across browsers

2. Body Styles
- Sets the default font and background
- Adds padding around the entire page

3. Component Styles
- Each section (header, nav, container, etc.) has its own style block
- Properties are grouped logically (color, spacing, layout)

Common CSS Properties Explained
- `background-color`: Sets background color
- `color`: Sets text color
- `padding`: Adds space inside elements
- `margin`: Adds space outside elements
- `border-radius`: Rounds corners
- `display`: Controls how elements are shown (block, flex, inline)
- `max-width`: Sets maximum width while allowing responsiveness

Helpful Resources

Documentation & Tutorials

- **MDN Web Docs** (developer.mozilla.org)
  - The most comprehensive and accurate documentation for HTML, CSS, and JavaScript
  - Includes interactive examples and browser compatibility information
  - Excellent tutorials for beginners to advanced developers
- **W3Schools** (w3schools.com)
  - Beginner-friendly tutorials with try-it-yourself editors
  - Simple explanations and lots of examples
  - Quick reference guides for HTML and CSS
- **CSS-Tricks** (css-tricks.com)
  - In-depth CSS tutorials and techniques
  - Excellent guides for Flexbox and Grid
  - Regular articles about modern web development

Framework & Tools

- **Bootstrap** (getbootstrap.com)
  - Popular CSS framework for building responsive websites
  - Extensive component library
  - Great documentation with examples
  - Ready-to-use templates

- **Tailwind CSS** (tailwindcss.com)
    - Modern utility-first CSS framework
    - Highly customizable
    - Excellent documentation with examples

Code Editors

- **Visual Studio Code** (code.visualstudio.com)
    - Free, powerful code editor
    - Great extensions for web development
    - Built-in live server capability

Practice & Learning

- **CodePen** (codepen.io)
    - Browser-based code editor
    - See and fork other people's code
    - Great for experimenting with HTML/CSS
- **freeCodeCamp** (freecodecamp.org)
    - Free, comprehensive web development curriculum
    - Interactive lessons and projects
    - Supportive community

Design Resources

- **Google Fonts** (fonts.google.com)
    - Free, high-quality web fonts
    - Easy to integrate into your projects
- **Font Awesome** (fontawesome.com)
    - Popular icon library
    - Free tier available with many icons

Color Tools

- **Coolors** (coolors.co)
    - Color scheme generator
    - Save and export color palettes
- **Color Hunt** (colorhunt.co)
    - Curated color palettes
    - Updated daily with new combinations

Validation Tools

- **W3C Validators** (validator.w3.org)
    - Official HTML and CSS validation
    - Helps find syntax errors and warnings

Community & Help

- **Stack Overflow** (stackoverflow.com)
    - Largest programming Q&A site
    - Search existing questions before asking new ones

Practice Exercises

1. Change the color scheme:
   - Try different background colors
   - Modify text colors
   - Experiment with button colors

2. Add new content:
   - Create a new article section
   - Add more navigation links
   - Include additional images

3. Modify layouts:
   - Change padding and margins
   - Adjust the container width
   - Try different border-radius values

Tips for Success

- Always save both files after making changes
- View your website by opening index.html in a web browser
- Refresh the browser to see changes
- Use browser developer tools (F12) to inspect elements
- Keep CSS organized by sections
- Test your website at different screen sizes

Common Issues and Solutions

- If styles aren't applying, check that:
  1. The CSS file is linked correctly in HTML
  2. File names match exactly
  3. Both files are in the same folder
  4. There are no typos in class names
  5. All curly braces and semicolons are present in CSS

Next Steps

After completing this basic website:

1. Learn about responsive design
2. Explore CSS flexbox and grid
3. Add interactive elements with JavaScript
4. Try using web fonts